



2025

V22

BOOKLET NO.

160035

Medical Science
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

P.T.O.

**SECTION – A**

- Q1.** Answer the following in about **150 words each** : **(5×10=50)**
- Q1.** (a) Describe hip joint under following heads : **(2+4+4=10)**
- i) Articular surfaces.
 - ii) Relations with suitable diagram.
 - iii) Applied aspects.
- Q1.** (b) Describe the formation, secretion, regulation and functions of the hormones of posterior pituitary. **10**
- Q1.** (c) Classify liver function tests. Describe the tests to assess synthetic functions of liver. **10**
- Q1.** (d) Define inflammation. Mention cardinal signs and types of inflammation and discuss chemical mediators of inflammation. **10**
- Q1.** (e) Describe phases of antibody production. Tabulate any six differences between humoral and cell mediated immunity. **10**
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- Q2.** (a) Describe uterus under following heads : **(5×4=20)**
- i) Gross Anatomy.
 - ii) Relations.
 - iii) Enumerate supports.
 - iv) Role of primary supports.
 - v) Clinical Anatomy.
- Q2.** (b) Describe the functions of cerebellum in overall motor control. **20**
- Q2.** (c) Describe the sources, biochemical functions and deficiency disorders of Vitamin B12. **10**
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- Q3.** (a) Define leukemia. Give WHO classification of acute myeloid leukemia. Describe in detail clinical features and lab diagnosis of acute myeloid leukemia. **20**
- Q3.** (b) Attempt the following : **(2×10=20)**
- i) Describe laboratory diagnosis of meningococcal meningitis. Enumerate meningococcal vaccines.
 - ii) Discuss serological tests used for diagnosis of enteric fever. Mention two categories of drug resistance in Salmonella Typhi.
- Q3.** (c) Define Poison. Classify Poisons according to their mode of action and as per criminal law. Add a note on Vitriolage. **10**
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- Q4.** (a) A 40 years male presents with polyphagia, polyuria and polydypsia since last 2 months. **20**
- i) What is your probable diagnosis ?
 - ii) Give its classification.
 - iii) Describe different investigations in detail to diagnose him.
 - iv) Write down the complications.
- Q4.** (b) Write the following with respect to malaria : **20**
- i) Name malarial parasites prevalent in India.
 - ii) Explain relapse and recrudescence.
 - iii) Give laboratory diagnosis of malaria.
- Q4.** (c) Explain the mechanism of action of methotrexate, an anticancer drug. Write its four side effects. **10**
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SECTION – B

- Q5.** Answer the following in about **150** words **each** : **(5×10=50)**
- Q5.** (a) Write mechanism of action of Metformin as an antidiabetic drug. What are side effects of Metformin ? **10**
- Q5.** (b) Define burns with its characteristic features. Enumerate varieties of burns. Describe Wallace Rule of Nine in adults. What are the causes of death in case of burns within 48 hours and after 48 hours ? **(2+3+2+3=10)**
- Q5.** (c) Describe the structure and functions of placenta. Add a note on human chorionic gonadotropin. **10**
- Q5.** (d) Classify lung tumours. Give gross and microscopic features of bronchogenic carcinoma. **10**
- Q5.** (e) Describe life cycle and laboratory diagnosis of *Giardia duodenalis*. **10**
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- Q6.** (a) Describe mammary gland under following heads : **(4×5=20)**
- i) Gross Anatomy.
 - ii) Blood supply.
 - iii) Lymphatic drainage.
 - iv) Clinical Anatomy.
- Q6.** (b) Discuss the role of kidney in long term control of blood pressure with special mention of role of the Renin-Angiotensin systems. **20**
- Q6.** (c) Describe Restriction Fragment Length Polymorphism (RFLP) and its clinical applications. **10**
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- Q7.** (a) Define cirrhosis. Give its classification. Describe etiopathogenesis, morphological features and complications of alcoholic liver disease. **20**
- Q7.** (b) Attempt the following : **(2×10=20)**
- i) Draw labelled diagram of Human Immunodeficiency Virus. Enumerate situations for HIV screening. Discuss important tests used in monitoring course of HIV infection.
 - ii) Describe morphology and antigenic types of polio virus. Add note on pulse polio immunization programme in India.
- Q7.** (c) Define Ophitoxaemia. Enumerate common poisonous snakes found in India. Differentiate between poisonous and non-poisonous snakes. Add a note on polyvalent Anti Snake Venom (ASV). **10**
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- Q8.** (a) What is Rheumatic fever ? Describe etiopathogenesis, clinical features and pathological changes in heart in case of rheumatic heart disease. **20**
- Q8.** (b) Attempt the following : **(2×10=20)**
- i) Describe clinical manifestations and laboratory diagnosis of mucocutaneous candidiasis.
 - ii) Mention clinical varieties of pulmonary aspergillosis. Discuss laboratory diagnosis of aspergillosis.
- Q8.** (c) What is the mechanism of action of chloroquine when used in malaria and rheumatoid arthritis ? Write its four side effects. **10**
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